

A three-dimensional heat transfer and thermal cracking model considering the effect of cracks on heat transfer

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Summary

A simple three-dimensional heat transfer model is developed to consider the hindering effect of cracks on heat transfer. The 3D heat transfer model can also be applied to numerical methods such as the combined finite-discrete element method (FDEM), discrete element method (DEM), discontinuous deformation analysis (DDA), the numerical manifold method (NMM), and the finite element method (FEM) to construct thermo-mechanical coupling models that allow these methods to solve thermal cracking problems and dynamically consider the hindering effect of cracks on heat transfer. In the 3D heat transfer model, the continuous-discontinuous medium is discretized into independent tetrahedral elements, and joint elements are inserted between adjacent tetrahedral elements. Heat transfer calculations for continuous-discontinuous media are converted to heat conduction in tetrahedral elements and the heat exchange between the adjacent tetrahedral elements through the joint element. If the joint element between adjacent tetrahedral elements breaks (ie, a crack generates), the heat exchange coefficient of the joint element is reduced to account for the hindering effect of cracks on heat conduction. Then the model and the FDEM are combined to build a thermo-mechanical coupling model to simulate thermal cracking. The thermally induced deformation, stress, and cracking are investigated by the thermo-mechanical coupling model, and the numerical results are compared with analytical solutions or experimental results. The 3D heat transfer model and thermo-mechanical model can provide a powerful tool for simulating heat transfer and thermal cracking in a continuous-discontinuous medium.

KEYWORDS

FDEM, heat conduction, thermal cracking, thermal stress, thermo-mechanical coupling model

1 | INTRODUCTION

Under the effect of temperature and a temperature gradient, a local thermal stress concentration will occur between mineral particles because of the anisotropy and heterogeneity of the thermal expansion of the rock mineral particles. If the stress concentration exceeds the tensile strength of the rock, thermal cracking will occur, and a fracture network will eventually form. Unlike hydraulic fracturing, which often generates cracks that extend along a certain direction,¹

thermal cracking creates a fracture network, which can increase both the fracture length and fracture density simultaneously. Therefore, thermal cracking has significant application prospects for increasing the permeability of reservoirs.

To understand the thermal cracking mechanism of rocks, many scholars use numerical models to study thermal cracking behavior. For example, Tang et al² developed a mesostructure-based damage model based on rock failure process analysis (RFPA)³ to simulate the thermal cracking behavior of ceramics. Jiang et al⁴ and Li et al⁵ investigated the formation mechanism of crack patterns in ceramics subjected to thermal shock based on the finite element method (FEM). Zhang et al⁶ studied the thermal shock analysis of 2D cracked solids using the numerical manifold method (NMM) and precise time integration. Bouhala et al⁷ investigated the thermal and thermo-mechanical influences on crack propagation using an extended meshfree method. D'Antuono and Morandini⁸ proposed a weakly coupled peridynamic formulation that can effectively simulate the occurrence and propagation of cracks due to extreme thermal loading. Wang et al⁹ proposed a thermo-mechanical coupling model based on the pre-dynamics method to simulate thermal cracking. Jiao et al¹⁰ proposed a two-dimensional thermo-mechanical coupling model based on discontinuous deformation analysis (DDA) to simulate rock fractures caused by thermal stress. Tomac and Gutierrez¹¹ simulated coupled forced heat convection and heat conduction using the discrete element method (DEM). Wanne and Young¹² and Xia et al¹³ simulated the thermal cracking of rock using particle flow code (PFC). André et al¹⁴ proposed a discrete element thermo-mechanical model of diffuse damage induced by thermal expansion mismatch of two-phase materials. Feng et al¹⁵ proposed a novel discrete thermal element (DTEM) method for modeling heat transfer between particles in two-dimensional situations. The above heat transfer and thermo-mechanical coupling models are mainly based on particle DEM. There are only few heat transfer and thermo-mechanical coupling models based on block DEM, mainly the heat transfer and thermo-mechanical coupling models of UDEC and 3DEC.^{16,17} However, these models based on block DEM cannot consider the effect of cracks on heat transfer and cannot consider thermal cracking.

To simulate the transition from a continuum to a discontinuum, the combined finite-discrete element method (FDEM) was proposed¹⁸⁻²¹ and an open source Y-code has been developed.¹⁹ The method is quite suitable for the simulation of rock fractures.²²⁻³¹ In recent years, FDEM has gained more and more attention. It has been widely used in various fields, such as for studying slopes,³² tunnels,³³ hydraulic fracturing,^{25,26,31,34,35} rock fracture mechanisms,²¹ and geothermal mining.²⁶ However, the FDEM could only be used to perform pure mechanical fracture calculations initially. Therefore, some hydro-mechanical coupled models based on the FDEM were developed for simulating hydraulic fracturing, which can be used to solve the fluid-driven fracturing of media with arbitrary complex fracture networks.³²⁻³⁹ Later, a two-dimensional thermo-mechanical coupling model based on the FDEM was also developed to simulate thermal cracking.⁴⁰ However, this thermo-mechanical coupling model takes the discontinuous medium as a continuum when conducting heat conduction calculations, so it cannot consider the hindering effect of cracks on heat transfer.

Therefore, in this paper, we propose a new, concise, three-dimensional heat transfer model for a continuous-discontinuous medium. In the 3D heat transfer model, the hindering effect of cracks on the subsequent heat transfer can be dynamically considered during the thermal cracking simulation. Then the 3D heat transfer model is combined with the mechanical fracture calculation of the FDEM to build a thermo-mechanical coupling model for simulating thermal cracking.

2 | FUNDAMENTALS OF FDEM

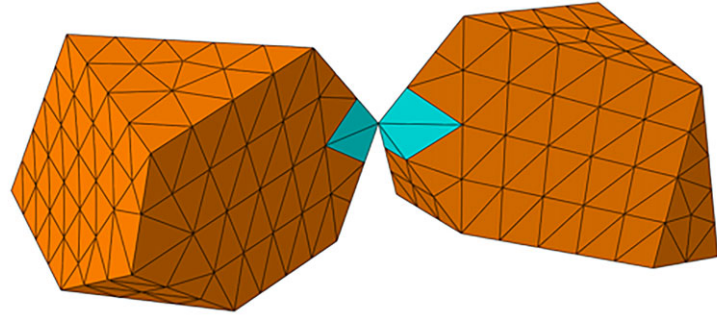
The mechanical fracture calculation of the thermal cracking model in this paper is based on the FDEM. The basic idea of the FDEM is that the continuous-discontinuous medium is divided into tetrahedral elements and joint elements with zero initial thickness are inserted in the adjacent tetrahedral elements, as shown in Figure 1A. In FDEM, the deformation of the continuum is simulated by the constant-strain tetrahedral element and the joint element with bonding effect while the crack initiation and propagation are simulated by the breaking of the joint element.

2.1 | System equation of FDEM

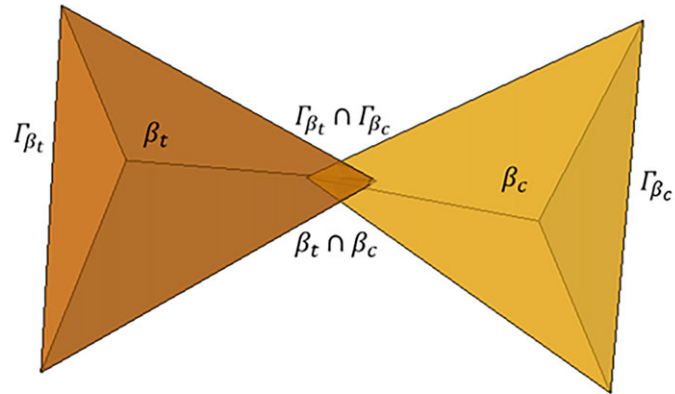
The system equation of FDEM is essentially the same as that of DEM.^{16,17,41,42} That is, according to Newton's second law, the dynamic equation of the nodes is given by

$$\mathbf{M}\ddot{\mathbf{x}} + \mathbf{C}\dot{\mathbf{x}} = \mathbf{F}(\mathbf{x}) \quad (1)$$

where $\mathbf{M}$ and $\mathbf{C}$ are the mass matrix and damping diagonal matrix of all nodes in the system. $\mathbf{F}(\mathbf{x})$ denotes the unbalanced nodal force vector, which includes nodal force $\mathbf{F}_c$ caused by the contact force, the nodal force $\mathbf{F}_d$ caused by the



(A) Discretization of blocks and the transformation of contact force



(B) Contact force between two tetrahedral elements

FIGURE 1 Contact force calculation
[Colour figure can be viewed at
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deformation of the tetrahedral element, the nodal force $\mathbf{F}_e$ of the external load distributed to the node, and the nodal force $\mathbf{F}_j$ caused by the deformation of the joint element. The damping matrix $\mathbf{C}$ is used to consume the kinetic energy of the system and to solve the static problem with the dynamic relaxation method. The damping matrix $\mathbf{C}$ is given by

$$\mathbf{C} = \mu \mathbf{I}, \quad (2)$$

where μ is the damping coefficient and $\mathbf{I}$ is the unit matrix. According to the single-degree-of-freedom particle spring system, the critical damping coefficient is given by

$$\mu = 2h\sqrt{\rho E}, \quad (3)$$

where h is the length of the element, ρ is the density, and E is the elastic modulus. The critical damping coefficient is used to consume the kinetic energy of the system at the fastest speed.

Equation 1 is solved in the same way as DEM, that is, the Euler method is used to solve the dynamic equation of the nodes. In each time step, the velocity and coordinates of the nodes are updated as the following:

$$\begin{aligned} \mathbf{v}_i^{(t+\Delta t)} &= \mathbf{v}_i^{(t)} + \sum \mathbf{F}_i^{(t)} \frac{\Delta t}{m_n} \\ \mathbf{x}_i^{(t+\Delta t)} &= \mathbf{x}_i^{(t)} + \mathbf{v}_i^{(t)} \Delta t, \end{aligned} \quad (4)$$

where $\mathbf{F}_i^{(t)}$ is the total nodal force, Δt is the time step, and m_n is the nodal mass, which is equal to one quarter of the tetrahedral element mass.

Next, we describe the constitutive of the tetrahedral element and the nodal force caused by the deformation of tetrahedral element, contact force and the nodal force caused by the contact force, and the constitutive of joint element and nodal force caused by the deformation of the joint element.

2.2 | Tetrahedral element constitutive and nodal force caused by tetrahedral element deformation

2.2.1 | Tetrahedral element constitutive

The stress-strain relationship of tetrahedral elements in FDEM satisfies the following equation

$$\mathbf{T} = \frac{1}{\sqrt{|\det \mathbf{F}|}} \left[\frac{E}{1+\nu} \mathbf{E}_d + \frac{E}{1-\nu} \mathbf{E}_s + 2\mu \mathbf{D} \right], \quad (5)$$

where $\mathbf{T}$ is the stress tensor, $\mathbf{F}$ is the deformation gradient, E is the elastic modulus, ν is the Poisson ratio, $\mathbf{E}_d$ is the Green strain tensor, $\mathbf{E}_s$ is the St Venant strain tensor, μ is the damping coefficient, $\mathbf{D}$ is the strain rate tensor. $\mathbf{F}$, $\mathbf{E}_d$, $\mathbf{E}_s$, and $\mathbf{D}$ can be obtained according to the current coordinates, initial coordinates, current velocity, and initial velocity of the four nodes of a tetrahedral element.

2.2.2 | Nodal force caused by deformation of the tetrahedral element

The nodal force caused by deformation of the tetrahedral element can be calculated according to the following equation:

$$\mathbf{f}_n = \frac{1}{3} \mathbf{T} \mathbf{n} S = \frac{1}{3} \begin{bmatrix} \sigma_{xx} & \sigma_{xy} & \sigma_{xz} \\ \sigma_{yx} & \sigma_{yy} & \sigma_{yz} \\ \sigma_{zx} & \sigma_{zy} & \sigma_{zz} \end{bmatrix} \begin{bmatrix} n_x \\ n_y \\ n_z \end{bmatrix} S, \quad (6)$$

where $\mathbf{n}$ is the outer normal unit vector of the tetrahedral element surface and S is the area of the tetrahedral element surface.

2.3 | Contact force and nodal force caused by the contact force

As shown in Figure 1A, the two blocks are divided into tetrahedral elements in FDEM. Therefore, the calculation of the contact force between the blocks is converted into the contact force calculation between the tetrahedral elements, as shown in Figure 1B. First, the FDEM adopts the NBS⁴³ algorithm proposed by Munjiza to find the contact pair. If two tetrahedral elements have been found in contact with each other, an equal but with opposite direction contact force is applied to the two tetrahedral elements to prevent excessive embedding. The contact force includes a normal contact force and a tangential contact force.

2.3.1 | Normal contact force

The normal contact force calculation is shown in Figure 1B. The two tetrahedral elements in one contact pair are called the contactor and the target, respectively. The contact force caused by the contactor embedded in the target with volume dV is given by¹⁹

$$d\mathbf{f}_{n1} = -p_n(\mathbf{grad} \varphi_t(P_t)) dV, \quad (7)$$

where p_n is the normal penalty value, P_t is the point of the target β_t , and $\varphi_t(P_t)$ is the potential of point P_t in the target β_t .

Similarly, the contact force caused by the target β_t embedded in the contactor β_c with volume dV is given by

$$d\mathbf{f}_{n2} = p_n(\mathbf{grad} \varphi_c(P_c)) dV. \quad (8)$$

Equation 8 does not have a negative sign because the contact force directions of Equation 8 are just opposite to that of Equation 7.

Then the total contact force caused by the overlapping volume dV (target to contactor) is given by

$$d\mathbf{f}_n = d\mathbf{f}_{n1} + d\mathbf{f}_{n2} = p_n[\mathbf{grad} \varphi_c(P_c) - \mathbf{grad} \varphi_t(P_t)] dV. \quad (9)$$

By integration of Equation 9 over the entire overlapping volume V , the total contact force between the two tetrahedral elements is given by

$$\mathbf{f}_n = \int_{V=\beta_{tr}\beta_c} p_n [\mathbf{grad} \varphi_c(P_c) - \mathbf{grad} \varphi_t(P_t)] dV. \quad (10)$$

Equation 10 can be written as an integration over the outer surface of the overlapping zone according to the Gaussian formula

$$\mathbf{f}_n = p_n \int_{S_{\beta_{tr}\beta_c}} \mathbf{n}(\varphi_c(P_c) - \varphi_t(P_t)) dS, \quad (11)$$

where $\mathbf{n}$ is the outer normal unit vector of the surface of the overlapping zone V .

Since the individual discrete blocks are discretized into tetrahedral finite element meshes in FDEM, the discrete blocks can be represented by the union of those tetrahedral elements

$$\left. \begin{aligned} \beta_c &= \beta_{c_1} \cup \beta_{c_2} \cdots \cup \beta_{c_i} \cdots \cup \beta_{c_n} \\ \beta_t &= \beta_{t_1} \cup \beta_{t_2} \cdots \cup \beta_{t_j} \cdots \cup \beta_{t_m} \end{aligned} \right\}. \quad (12)$$

The volume integral of the overlapping zone of the two contact discrete blocks can then be expressed by the summation of the integration over the overlapping zone of the tetrahedral elements.¹⁹

$$\mathbf{f}_n = \sum_{i=1}^n \sum_{j=1}^m \int_{\beta_{c_i} \cap \beta_{t_j}} p_n [\mathbf{grad} \varphi_{c_i}(P_{c_j}) - \mathbf{grad} \varphi_{t_j}(P_{t_j})] dV. \quad (13)$$

The integral of the volume of the overlapping zone is converted to the integral of the surface of the overlapping zone by the Gaussian formula

$$\mathbf{f}_n = \sum_{i=1}^n \sum_{j=1}^m p_n \int_{S_{\beta_{c_i} \cap \beta_{t_j}}} \mathbf{n}(\varphi_c - \varphi_t) dS. \quad (14)$$

Munjiza et al^{19,44} pointed out that the potential on the boundary of the tetrahedral element should be constant to ensure the work done by the contact force is zero. Accordingly, Munjiza defines the following potential function

$$\varphi(P) = \min\{4V_1/V, 4V_2/V, 4V_3/V, 4V_4/V\}, \quad (15)$$

where V_1 , V_2 , V_3 , and V_4 are the volume of the subtetrahedron that is formed by the point P and the four surfaces of the tetrahedral element, and V is the total volume of the tetrahedral element. The potential on the boundary of the tetrahedral element is 0, and the potential at the center of the tetrahedral element is 1. In order to make it easier to understand, the tetrahedral element is divided into four subtetrahedrons by the center point k of the tetrahedral element and the four surfaces of the tetrahedral element, as shown in Figure 2. If point P is located in the subtetrahedron $ijkp$, the potential¹⁴ of point P can be further written as

$$\varphi(P) = \frac{V_{ijkp}}{4V_{ijkl}}. \quad (16)$$

The potential here is similar to the concept of gravitational potential. The advantage of using potential to define contact force is that the work done by the contact force is only related to the starting and ending position but has nothing to do with the specific path. Since the potential at the boundary of the tetrahedral element is 0, the total work done by the contact force is always zero. Thus, the system energy is conserved.

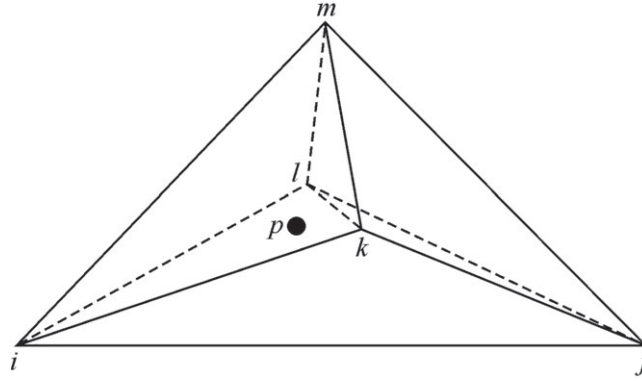


FIGURE 2 Subdivision of the target tetrahedral element

2.3.2 | Tangential contact force

After obtaining the normal contact force and the acting point of the normal contact force, we calculate the tangential contact force according to the classical Coulomb friction

$$\mathbf{f}_t^t = \mathbf{f}_t^{t-\Delta t} - p_t \Delta \mathbf{u}_s S_B, \quad (17)$$

where $\mathbf{f}_t^t$ is the friction at the current time step, $\mathbf{f}_t^{t-\Delta t}$ is the trial friction at the last time step, and p_t is the tangential penalty parameter. $\Delta \mathbf{u}_s$ is the tangential relative displacement increment at one time step. S_B is the area of the contact surface. According to Equation 17, if $|\mathbf{f}_t^t| \leq |\mu \mathbf{f}_n|$, then

$$\mathbf{f}_t = \mathbf{f}_t^t. \quad (18)$$

According to Equation 17, if $|\mathbf{f}_t^t| > |\mu \mathbf{f}_n|$, then

$$\mathbf{f}_t = \frac{\mathbf{f}_t^t}{|\mathbf{f}_t^t|} |\mu \mathbf{f}_n|. \quad (19)$$

Thus, the tangential contact force can be obtained.

The above normal contact force and tangential contact force are distributed to the nodes of the two tetrahedral elements in contact, that is, the nodal force caused by the contact force is obtained.

2.4 | Joint element constitutive and nodal force caused by joint element deformation

As shown in Figure 3, a six-node joint element with bonding effect is inserted in the adjacent tetrahedral element. The combine single and smeared crack model¹⁰ are used as the constitutive and failure criterion of the joint element. The crack initiation and propagation in continuum are simulated through the joint element breaking. Since the joint element shares the same nodes with the adjacent two tetrahedral elements, crack extends only along the boundary of the tetrahedral elements. Thus, the crack initiation and propagation in FDEM have a mesh dependence, but as the mesh density increases, this mesh dependence will gradually weaken.^{22,45}

As shown in Figure 3A, the coordinate vector of the top three nodes of the joint element is $\mathbf{U}_i (i = 1, 2, 3)$, whereas the coordinate vector of the three nodes on the bottom surface of the joint element is $\mathbf{L}_i (i = 1, 2, 3)$. The three nodes at the middle section of the joint element are given by

$$\mathbf{M}_i = \frac{\mathbf{U}_i + \mathbf{L}_i}{2} \quad (20)$$

Initially, the top surface overlaps with the bottom surface of the joint element. However, any two overlapping points $\mathbf{p}$ and $\mathbf{p}'$ at the top and bottom surfaces are not coincide when the joint element is open. The open vector δ of the joint element at one point can be expressed as

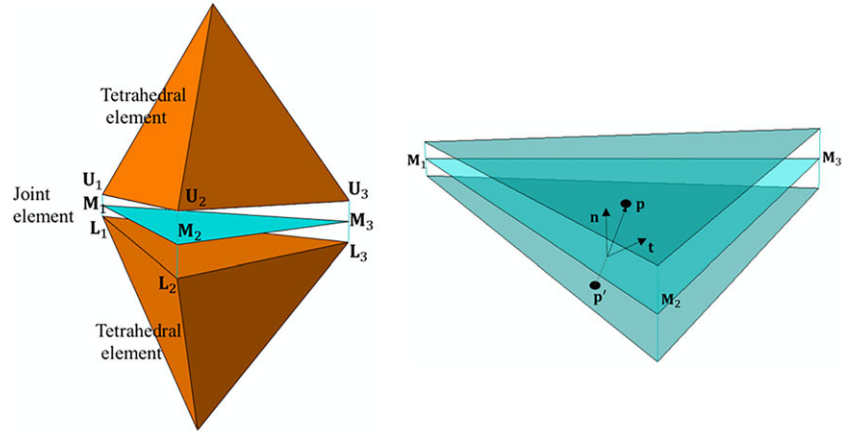


FIGURE 3 Schematic diagram of joint element opening amount calculation.³⁷ A, Connection of the joint element and tetrahedral elements. B, Schematic diagram of opening amount calculation of the joint element. [Colour figure can be viewed at wileyonlinelibrary.com]

(A) Connection of the joint element and tetrahedral elements. (B) Schematic diagram of opening amount calculation of the joint element.

$$\delta = \mathbf{p} - \mathbf{p}' \quad (21)$$

The open vector can be decomposed into

$$\delta = o\mathbf{n} + s\mathbf{t} \quad (22)$$

where o and s are the normal component and the tangential component of the open vector, respectively, namely, the normal opening amount o and the tangential slippage amount s ; $\mathbf{n}$ is the normal unit vector of the middle section of the joint element; and $\mathbf{t}$ is the unit direction vector of the projection direction of the open vector on the middle section. These two vectors can be expressed as

$$\mathbf{n} = \frac{(\mathbf{M}_2 - \mathbf{M}_1) \times (\mathbf{M}_3 - \mathbf{M}_1)}{|(\mathbf{M}_2 - \mathbf{M}_1) \times (\mathbf{M}_3 - \mathbf{M}_1)|} \quad (23)$$

$$\mathbf{t} = \delta - (\delta \cdot \mathbf{n})\mathbf{n}$$

Thus, the normal opening amount o and tangential slippage amount s can be expressed as

$$o = \delta \cdot \mathbf{n} \quad (24)$$

$$s = \delta \cdot \mathbf{t}$$

Similarly, the stress vector of the joint element can be decomposed into

$$\mathbf{T} = \sigma\mathbf{n} + \tau\mathbf{t} \quad (25)$$

where σ is the normal stress component and τ is the tangential stress component.

Next, we describe the relationship between the normal stress component σ and the normal opening amount o , and between the tangential stress component τ and the tangential slippage amount s .

Theoretically, the normal opening amount of a point should be zero (ie, $o = o_p = 0$) when the normal stress at the point is equal to or less than the tensile strength (ie, $\sigma = f_s$). That is, there should be no crack opening before the normal stress at one point reaches the tensile strength. However, the opening amount of a joint element will never be zero because the two adjacent tetrahedral elements do not share the same nodes. In order to ensure that the opening amount between the tetrahedral elements is sufficiently small, a penalty function method is used, which apply a normal constraint force on the two tetrahedral elements.

When $0 < o \leq o_p$, the normal stress of a joint element can be expressed as

$$\sigma = \left[2o/o_p - (o/o_p)^2 \right] f_t, \quad (26)$$

where $o_p = 2hf_t/p_n$ is the normal opening amount when the normal stress σ is equal to the tensile strength f_t , h is the edge length of the joint element, and p_n is the normal penalty parameter. When it tends to infinity, $o_p = \lim_{p_n \rightarrow +\infty} 2hf_t/p_n = 0$. That is, if the normal penalty parameter is large enough, the normal opening amount will tend to zero before the normal stress reaches the tensile strength.

When $o_p < o \leq (o_p + o_t)$ ($(o_p + o_t)$ is the critical normal opening amount when the normal stress decreases to zero), the normal stress will be gradually reduced as the opening amount increases. The normal stress can be expressed as

$$\sigma = z f_t, \quad (27)$$

where z is the expression representing the strain softening, it is given by

$$z = \left[1 - \frac{a+b-1}{a+b} e^{d\{a+cb/[(a+b)(1-a-b)]\}} \right] \cdot [a(1-d) + b(1-d)^c], \quad (28)$$

where a , b , c are obtained by fitting the experimental curve of the material under tension and d is a dimensionless quantity to characterize the crack opening. It can be given by

$$d = \begin{cases} \sqrt{\left(\frac{o-o_p}{o_t}\right)^2 + \left(\frac{|s|-s_p}{s_t}\right)^2}, & (o_p + o_t) > o > o_p \&\& (s_p + s_t) > s > s_p \\ \frac{o-o_p}{o_t}, & (o_p + o_t) > o > o_p \&\& |s| \leq s_p \\ \frac{|s|-s_p}{s_t}, & o \leq o_p \&\& (s_p + s_t) > |s| > s_p \\ 1, & o \geq (o_p + o_t) \&\& |s| \geq (s_p + s_t) \\ 0, & \text{otherwise,} \end{cases} \quad (29)$$

where s is the tangential slippage, s_p is the tangential slippage amount when the tangential stress is equal to the shear strength, and $(s_p + s_t)$ is the critical tangential slippage amount when the tangential stress decreases to zero. If $d \geq 1$, the joint element is broken and the two tetrahedral elements connected by the joint element are completely disconnected.

It should be noted that the critical normal opening amount and the critical tangential slippage amount depend on the tensile and shear strength, and the mode I and mode II fracture energy release rates, Gf_I and Gf_{II} , are defined as

$$Gf_I = \int_{o_p}^{o_p+o_t} \sigma do, \quad (30)$$

$$Gf_{II} = \int_{s_p}^{s_p+s_t} \tau ds. \quad (31)$$

In addition, the joint element is in a compressed state when $o \leq 0$. In order to make the embedding amount between the two tetrahedral elements sufficiently small, a penalty function method is also used, which apply a normal repulsive force for the two tetrahedral elements. The normal stress can be expressed as

$$\sigma = 2 \frac{o}{o_p} f_t. \quad (32)$$

In this way, the complete relationship between normal stress σ and normal opening amount o is obtained

$$\sigma = \begin{cases} 2 \frac{o}{o_p} f_t, & o < 0 \\ \left[\frac{2o}{o_p} - \left(\frac{o}{o_p} \right)^2 \right] f_t, & 0 \leq o \leq o_p \\ z f_t, & o > o_p \end{cases}. \quad (33)$$

Similarly, the relationship between the tangential stress τ and the tangential slip s can also be obtained. The relationship between the normal stress and the normal opening amount, and between the tangential stress and the tangential slippage amount can be visually represented by Figure 4.

The above normal and tangential stresses are integrated over the joint element and distributed to the nodes connected by the joint elements, that is, the nodal forces caused by the deformation of the joint element are obtained.

3 | THREE-DIMENSIONAL HEAT TRANSFER MODEL CONSIDERING THE HINDERING EFFECT OF CRACKS ON HEAT TRANSFER

3.1 | Basic concepts

In the 3D heat transfer model, the continuation-discontinuous medium is divided into a tetrahedral finite element mesh, as shown in Figure 5A. Joint elements are inserted between adjacent tetrahedral elements. Thus, the original coplanar tetrahedral elements are separated, and the three shared vertices are separated into six vertices, as shown in Figure 5B. At the same time, the joint element where the crack is located is marked as broken. In this way, the heat

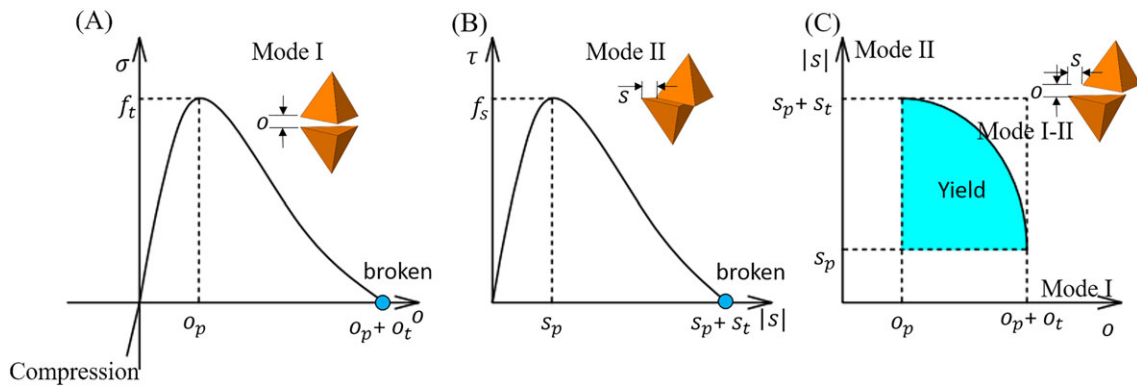


FIGURE 4 Fracture criteria of the joint element: A, mode I, tensile failure; B, mode II, shear failure; C, modes I-II, tensile-shear hybrid failure [Colour figure can be viewed at wileyonlinelibrary.com]

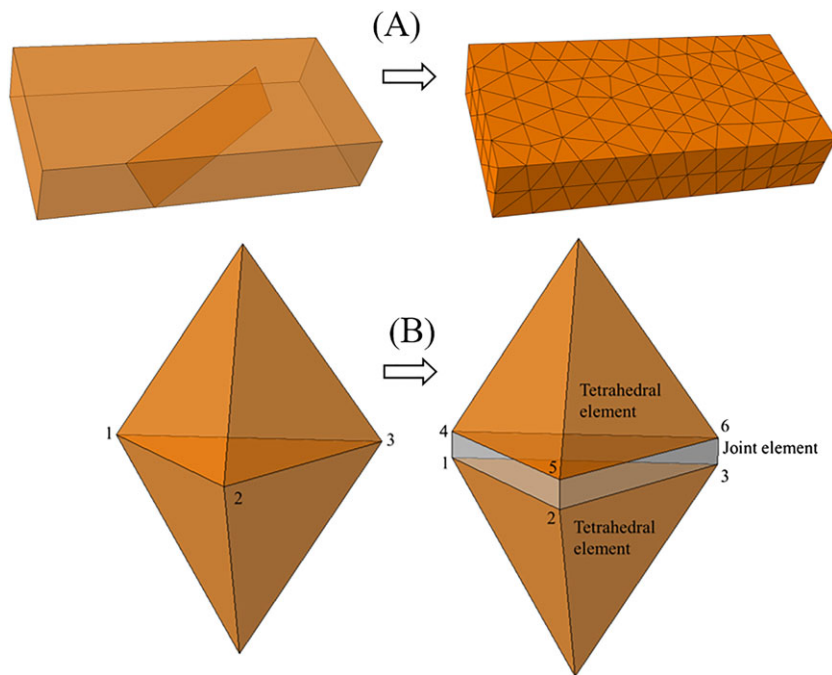


FIGURE 5 Computational mesh of the continuous-discontinuous medium and the topological connection between the tetrahedral element and joint element [Colour figure can be viewed at wileyonlinelibrary.com]

transfer in the continuous-discontinuous medium is converted into heat conduction in the tetrahedral elements and heat exchange between adjacent tetrahedral elements through the joint element.

The temperature distribution in the tetrahedral element is obtained through linear interpolation based on the temperatures at the four nodes of the tetrahedral element. The heat conduction in a single tetrahedral element is calculated according to the Fourier law of heat conduction; the heat exchange between adjacent tetrahedral elements is calculated according to the nodal temperature difference and the heat exchange coefficient of the joint element. For a broken joint element, the heat exchange coefficient is multiplied by a reduction factor to account for the hindering effect of cracks on heat transfer.

Thus, the 3D heat transfer model can be used to solve heat transfer problems in a continuous-discontinuous medium. Next, we describe the calculation of heat conduction in a tetrahedral element and the heat exchange between adjacent tetrahedral elements through the joint element.

3.2 | Heat transfer model

According to the Fourier law of heat conduction, the heat flow rate per unit area along the i direction can be expressed as

$$q_i = -k_{ij} \frac{\partial T}{\partial x_j} \quad (34)$$

where q_i is the heat flow rate, k_{ij} is the thermal conductivity coefficient tensor, and T is the temperature.

For any given mass M , the temperature change can be written as

$$\frac{\partial T}{\partial t} = \frac{Q_{total}}{C_p M} \quad (35)$$

where Q_{total} is the total heat flow into mass M per unit time, C_p is the specific heat, and M is the mass.

Based on the topological connection of the tetrahedral elements and the joint elements, a 3D heat transfer model is constructed as shown in Figure 6. The temperature of the tetrahedral element is represented by the temperatures at the

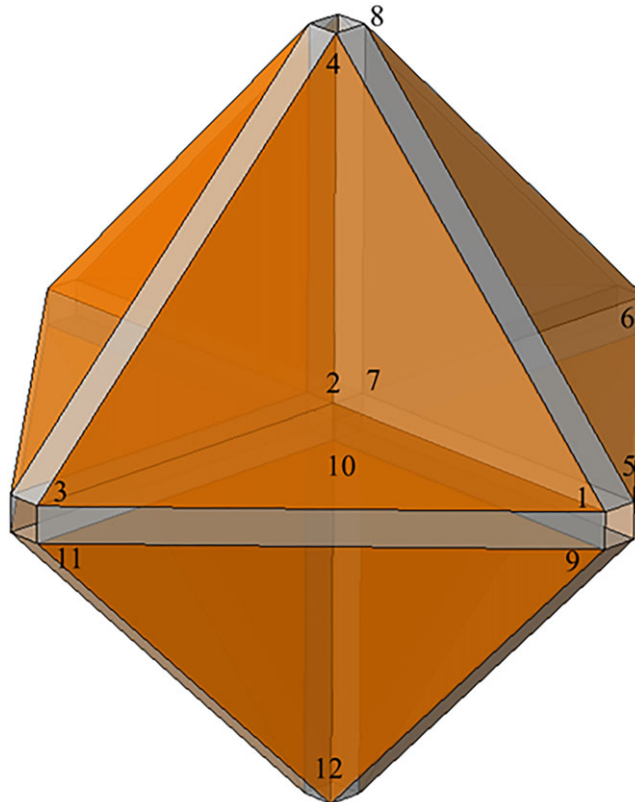


FIGURE 6 Three-dimensional heat transfer model in a medium without cracks [Colour figure can be viewed at wileyonlinelibrary.com]

four nodes of the tetrahedral element. Thus, the temperature of the entire continuous-discontinuous medium is characterized by the temperatures at the nodes of all the tetrahedral elements. Next, we introduce how to calculate the temperature field in the medium based on the topology connection shown in Figure 6.

Taking node 1 in Figure 6 as an example, the tetrahedral element at which node 1 is located is T1234, and the joint elements that connect to node 1 include J124578 and J1239'10'11. The other tetrahedral elements connected by the two joint elements include T9'10'11'12 and T5678. Since the temperature of node 1 may be different from those of nodes 2, 3, and 4 in tetrahedral element T1234, heat conduction may occur in tetrahedral element T1234. In addition, the temperature of node 1 may be different from those of nodes 5 and 9. Therefore, heat exchange between node 1 and node 5 or 9 may occur through joint elements J124578 and J1239'10'11.

3.2.1 | Heat conduction in tetrahedral elements and heat exchange between tetrahedral elements

Using tetrahedral element T1234, which directly connects to node 1, as an example, let the temperatures of these four nodes be T_i and the node coordinates be (x_i, y_i, z_i) (i is the node number, $i = 1, 2, 3, \text{ and } 4$). Assuming that the temperature distribution in the tetrahedral element obeys a linear distribution, the temperature gradient in the tetrahedral element is a constant and can be expressed as

$$\frac{\partial T}{\partial x_i} = \frac{1}{V} \int \frac{\partial T}{\partial x_i} dV. \quad (36)$$

According to the Gaussian divergence theorem, Equation 36 can be written as

$$\frac{\partial T}{\partial x_i} = \frac{1}{V} \int_S T n_i dS = -\frac{1}{3V} \sum_{l=1}^4 T^l n_i^{(l)} S^{(l)}, \quad (37)$$

where V is the volume of the tetrahedral element, $n_i^{(l)}$ is the outward normal unit vector of the triangular face l , T^l is the temperature of the node opposite to triangular face l , and $S^{(l)}$ is the area of triangular face l .

Substituting Equation 37 into Equation 34, the heat flow rate along the x , y , and z directions (q_x , q_y , and q_z) can be obtained.

Thus, the heat flow into node 1 per unit time can be calculated by

$$Q_{T1234 \rightarrow 1} = -\frac{q_i n_i^{(1)} S^{(1)}}{3}, \quad (38)$$

where $n_i^{(1)}$ is the outward normal unit vector of the triangular face opposite to node 1 and $S^{(1)}$ is the area of the triangular face opposite to node 1.

In this way, the heat flow into node 1 from tetrahedral element T1234 is obtained.

Tetrahedral element T1234 connects with tetrahedral elements T5678 and T9'10'11'12 through joint elements J124578 and J1239'10'11, respectively. Therefore, tetrahedral element T1234 exchanges heat with tetrahedral elements T5678 and T9'10'11'12 through joint elements J124578 and J1239'10'11, respectively. Taking tetrahedral elements T1234 and T5678 as an example, we draw a schematic diagram shown in Figure 7. Let the temperatures of nodes 1, 2, 3, 4, 5, 6, 7, and 8 be $T_1, T_2, T_3, T_4, T_5, T_6, T_7$, and T_8 . The heat exchange coefficient of the joint element is h_j . Then the heat exchange between the two tetrahedral elements T1234 and T5678 through joint element J124578 per unit time can be expressed by

$$Q = \frac{1}{3} h_j (T_5 + T_7 + T_8 - T_1 - T_2 - T_4) S = \frac{1}{3} h_j (T_5 - T_1) S + \frac{1}{3} h_j (T_7 - T_2) S + \frac{1}{3} h_j (T_8 - T_4) S \quad (39)$$

Thus, the heat flow rate (from tetrahedral element T5678 into tetrahedral element T1234 through joint element J124578) to node 1 is given by

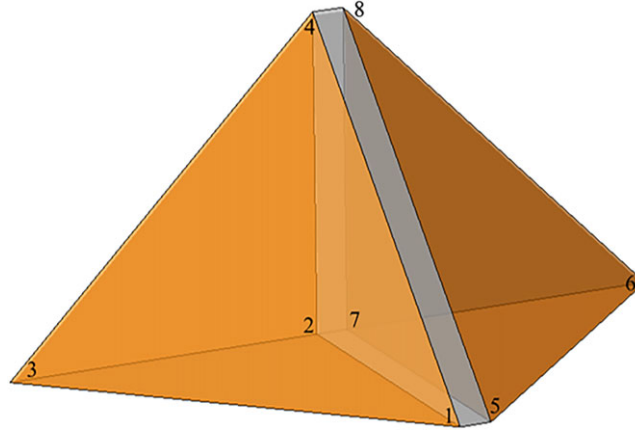


FIGURE 7 Heat exchange between the tetrahedral elements through the joint element [Colour figure can be viewed at wileyonlinelibrary.com]

$$Q_{T5678 \rightarrow 1} = \frac{1}{3} h_j (T_5 - T_1) S \quad (40)$$

Similar to Equation 40, the heat flow rate (from tetrahedral element T9'10'11'12 into tetrahedral element T1234 through joint element J1239'10'11) to node 1 is given by

$$Q_{T9'10'11'12 \rightarrow 1} = \frac{1}{3} h_j (T_9 - T_1) S \quad (41)$$

For continuous media, the joint element between the tetrahedral elements hinders heat transfer. To sufficiently reduce the thermal hindering effect of the joint element to meet the requirements of heat conduction in a continuous medium, the heat exchange coefficient of the joint element should take a very large value. The selection of the value of the heat exchange coefficient of the unbroken joint element will be discussed in detail in Section 4.1. According to the sensitivity analysis in Section 4.1, we find that if $h_j \geq 100 \frac{k}{L_e}$ (where k is the thermal conductivity coefficient of the tetrahedral element and L_e is the element size), then the numerical results will agree well with the analytical solutions of the heat conduction in a continuous medium.

If the continuum is divided by cracks, the cracks will act as a barrier to thermal transfer. For example, joint element J124578 in Figure 8 is marked as broken, differing from Figure 6. Still using node 1 as an example, the heat transfer from tetrahedral element T5678 into tetrahedral element T1234 through joint element J124578 should be reduced because joint element J124578 is broken (ie, joint element J124578 is a crack). That is, the heat flow rate to node 1 should be reduced. To express this hindering effect of cracks on heat transfer, we modify the heat flow rate (from tetrahedral T5678 into T1234 through the broken joint element) assigned to node 1 as

$$Q_{T5678 \rightarrow 1} = \frac{1}{3} h_c (T_5 - T_1) S = \frac{1}{3} (r_T h_j) (T_5 - T_1) S \quad (42)$$

where r_T is the reduction coefficient for the broken joint element ($0 < r_T < 1$). $r_T = 0$ means complete insulation, while $r_T = 1$ means that cracks do not have any effect on heat transfer.

If other joint elements break, the heat exchange coefficient of these joint elements also needs to be multiplied by the reduction factor r_T when calculating the nodal heat flow rate, as in Equation 40.

3.2.2 | Node temperature updating

Thus, the total heat flow rate into node 1 per unit time is given by

$$Q_{total} = Q_{T1234 \rightarrow 1} + Q_{T5678 \rightarrow 1} + Q_{T9'10'11'12 \rightarrow 1} \quad (43)$$

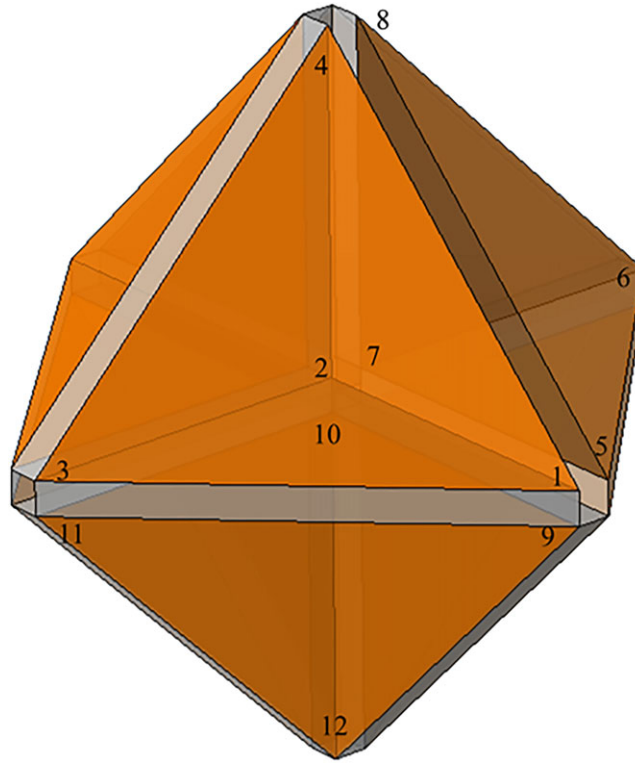


FIGURE 8 Heat transfer calculation in a medium with cracks [Colour figure can be viewed at wileyonlinelibrary.com]

Then, according to Equation 35, the temperature of node 1 at the next time step is given by

$$T_1^{t+\Delta t} = T_1^t + \frac{Q_{total}}{C_p M} \Delta t \quad (44)$$

where M is the nodal mass, which is equal to one-quarter of the mass of the tetrahedral element.

Thus, the temperature of node 1 at the next time step is updated. The temperatures of the other nodes can be updated similarly according to the above procedure. Then the temperature evolution of the entire solution domain is obtained.

Since an explicit algorithm is used in the heat transfer calculation, the time step needs to be less than the critical time to ensure the stability of the numerical calculation. The critical time step is given by¹⁷

$$\Delta t_c = \frac{(\Delta x)^2}{4\kappa \left[1 + \frac{h\Delta x}{2k} \right]} \quad (45)$$

where Δx is the size of the smallest element, h is the heat exchange coefficient, and κ is the thermal diffusivity ($k/\rho C_p$ when $k_x = k_y$).

3.2.3 | Determination of heat transfer reduction factor of crack

As shown in Figure 9, a strip contains a crack. The temperature on the left and right boundary of the strip are fixed as T_1 and T_2 , respectively. The thermal conductivity coefficient of the strip on both sides of the crack are k_1 and k_2 , and the lengths at the left and right of the crack are L_1 and L_2 . After the heat conduction is stabilized, the temperature difference ΔT between the left and right sides of the crack is measured by the sensor. Then, according to the Fourier heat conduction law,

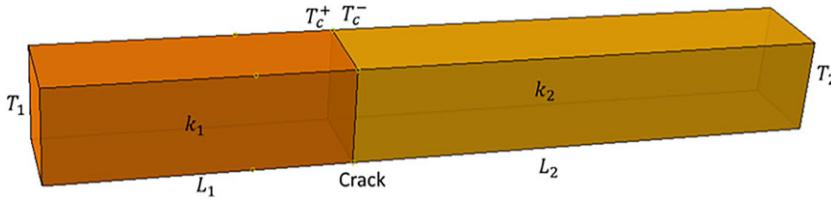


FIGURE 9 Determination of the heat transfer reduction factor at the crack [Colour figure can be viewed at wileyonlinelibrary.com]

$$\left. \begin{aligned} q &= h_c (T_c^+ - T_c^-) \\ q &= k_1 \frac{T_1 - T_c^+}{L_1} \\ q &= k_2 \frac{T_c^- - T_2}{L_2} \\ \Delta T &= T_c^+ - T_c^- \end{aligned} \right\} \quad (46)$$

According to the Equation 46, the relationship between the heat exchange coefficient of the crack and the temperature difference between left and right of the crack can be obtained as follows

$$h_c = \frac{k_1 k_2 (T_1 - T_2 - \Delta T)}{(k_1 L_2 + k_2 L_1) \Delta T} \quad (47)$$

In particular, if $k_1 = k_2 = k$ and $L_1 = L_2 = L$, the above equation can be rewritten as

$$h_c = \frac{k(T_1 - T_2 - \Delta T_c)}{2L\Delta T_c} \quad (48)$$

Then, according to Equations 48 and 42, we can obtain

$$r_T = \frac{k(T_1 - T_2 - \Delta T_c)}{2h_j L \Delta T_c} \quad (49)$$

Thus, we determine the reduction factor of heat exchange coefficient for the crack (ie, broken joint element in FDEM).

4 | THERMO-MECHANICAL COUPLING

4.1 | Calculation process

A temperature change causes a rock mass to expand or shrink. If the expansion or shrinkage deformation in a rock is constrained, then thermal stress is generated, and thermal cracking may occur, ie, cracks are generated. The generated cracks will in turn affect the heat transfer and thus affect the temperature field in the rock mass. The proposed 3D thermo-mechanical coupling model not only considers the thermal deformation and thermal stress caused by the temperature change but also considers the effect of crack initiation and propagation on the temperature field. The entire calculation process can be divided into four main steps: (1) The temperature distribution of the system is analyzed based on the 3D heat transfer model; (2) the thermal stress caused by the temperature change is calculated; (3) the temperature-induced thermal stress is applied to the system equation, the mechanical fracture calculation is performed, and the fracture network is updated; and (4) the heat exchange coefficient of the newly generated broken joint element is multiplied by a reduction factor r_T , which will be used as the input for the thermal transfer calculation at the next time step. According to above four steps, the thermo-mechanical coupling analysis is completed in a one-time step,

and the thermal cracking of rock can be simulated, as shown in Figure 10. The first step was introduced in Section 7. Next, steps 2, 3, and 4 will be presented.

4.2 | Thermal stress calculation

The thermal stress caused by the temperature change ΔT is given by

$$\Delta\sigma_{ij} = -3K\alpha\Delta T\delta_{ij}, \quad (50)$$

where $\Delta\sigma_{ij}$ is the thermal stress increment, K is the bulk modulus, and α is the thermal expansion coefficient.

Thus, the thermal stress under a given temperature field is obtained. Then the thermal stress is applied as a body load to the tetrahedral elements. The equivalent nodal force induced by the thermal stress can be express as

$$\mathbf{f}_n = -K^*\alpha\Delta T n_j S \delta_{ij}, \quad (51)$$

where n is the outward normal vector of the triangular face of the tetrahedral element and S is the area of the triangular face.

The nodal force is used as the load input for the mechanical fracture calculation of other numerical methods, such as the FEM, DEM, DDA,^{46,47} and NMM.⁴⁸⁻⁵⁰ Thus, thermal cracking of the rock mass can be simulated. The effect of cracks caused by thermal cracking on the heat transfer is taken into account by multiplying the heat exchange coefficient of the broken joint element by a suitable reduction factor r_T .

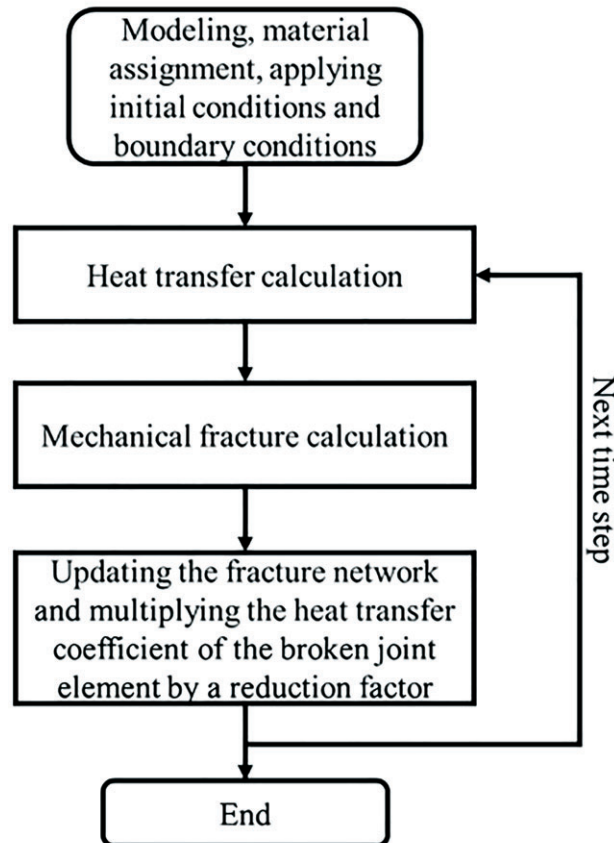


FIGURE 10 Thermo-mechanical coupling calculation process

5 | EXAMPLE VERIFICATION

5.1 | Transient heat conduction in a continuum

Take a block¹⁶ with length L , width W , and height H . The initial temperature of the block is T_2 . The top, bottom, front, and rear boundaries are thermally insulated, and the temperatures at the left and right boundaries of the block are T_1 and T_2 , respectively. The 3D heat transfer model is used to study the temperature evolution in the block.

The analytical solution to this problem is given by⁵¹

$$T(x, t) = T_1 + \frac{x}{L}(T_2 - T_1) + \frac{2}{\pi} \sum_{n=1}^{\infty} e^{-kn^2\pi t/L^2} \left(\frac{T_2 \cos(n\pi) - T_1}{n} \right) \sin \frac{n\pi x}{L}, \quad (52)$$

where T_1 is the temperature at the left boundary of the block, T_2 is the temperature at right boundary, t is time, x is the distance from the left boundary, $\kappa = k/(\rho C_p)$, k is the thermal conductivity coefficient, ρ is the density, and C_p is the specific heat.

Take $L = 0.1$ m, $W = H = 0.01$ m, $T_1 = 100^\circ\text{C}$, and $T_2 = 0^\circ\text{C}$. The block is divided into tetrahedral elements with an element size $L_e = 0.005$ m. The coordinates of all nodes are fixed regardless of the mechanical fracture calculation. The calculation parameters include the thermal conductivity coefficient $k = 4.2$ W/m $\cdot^\circ\text{C}$, the specific heat $C_p = 880$ J/kg $\cdot^\circ\text{C}$, and the density $\rho = 2500$ kg/m³.

Joint elements are inserted between adjacent tetrahedral elements, hindering the heat transfer and making the numerical results different from the analytical solution of heat conduction in the continuum. To reduce the hindering effect of the joint element, the heat exchange coefficient of the joint element should, theoretically, have an infinite value. However, according to Equation 45, the calculation time step should be infinitesimal when $h_j n$ is infinite, which is unacceptable for numerical calculations. Therefore, we hope to determine a suitable value for the heat exchange coefficient of the joint element that ensures a good agreement of the numerical result with the analytical solution and does not make the calculation time step extremely small.

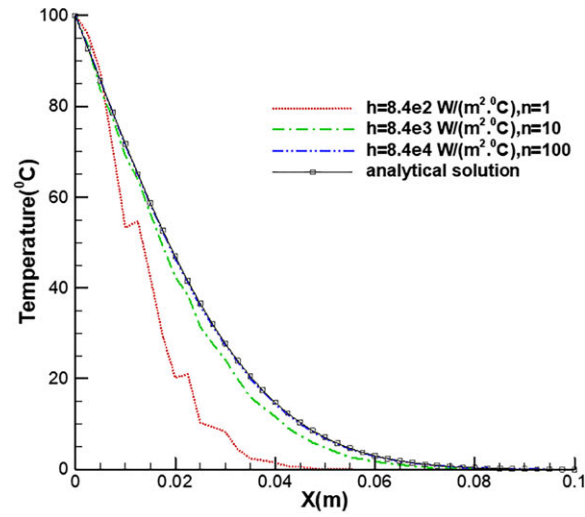
According to dimensional analysis, the dimension of k/L (k is the thermal conductivity coefficient and L is length) is consistent with the dimension of the heat exchange coefficient h_j . Here, we speculate that if h_j is much larger than $\frac{k}{L_e}$ (L_e is the element size), then the numerical results of the 3D heat transfer model will be consistent with the analytical solution of the heat conduction in the continuum. For ease of discussion, let $h_j = n \left(\frac{k}{L_e} \right)$, where n is a magnification factor. Next, we perform sensitivity analysis for the heat exchange coefficient of the joint element and study the effect of the thermal conductivity coefficient, element size, and element shape on the heat exchange coefficient of the joint element. Finally, the criteria for selecting h_j are determined.

5.1.1 | Effect of thermal conductivity coefficient k on heat exchange coefficient h_j of the joint element

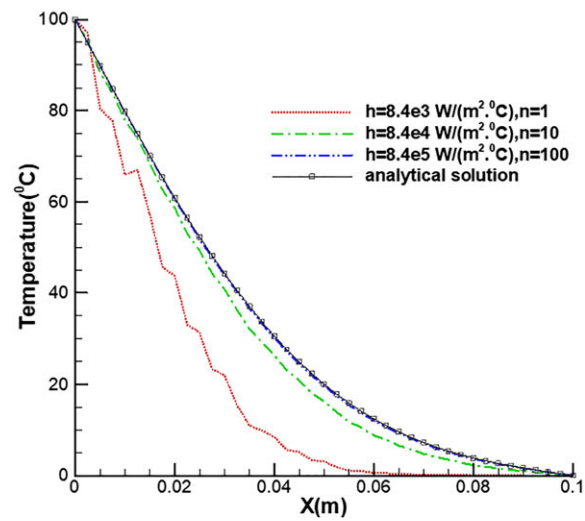
We keep the size of the tetrahedral element at 0.005 m and then change the thermal conductivity coefficient of the tetrahedral element to $k = 4.2, 42$, and 420 W/m $\cdot^\circ\text{C}$. Then the 3D heat transfer model is used to obtain the temperature distribution under different thermal conductivity coefficients. As shown in Figure 11A-C, when k increases from 4.2 to 420 W/m $\cdot^\circ\text{C}$, the heat exchange coefficient h_j of the joint element needs to increase from 8.4×10^2 W/(m² $\cdot^\circ\text{C}$) to 8.4×10^4 W/(m² $\cdot^\circ\text{C}$) to make the temperature distribution obtained from the 3D heat transfer model consistent with the analytical solution. However, we find that the amplification factor n does not need to increase with an increase of the heat conductivity coefficient k . As long as $n \geq 100$, the numerical result of the heat transfer model will be consistent with the analytical solution.

5.1.2 | Effect of the element size on the heat exchange coefficient of the joint element

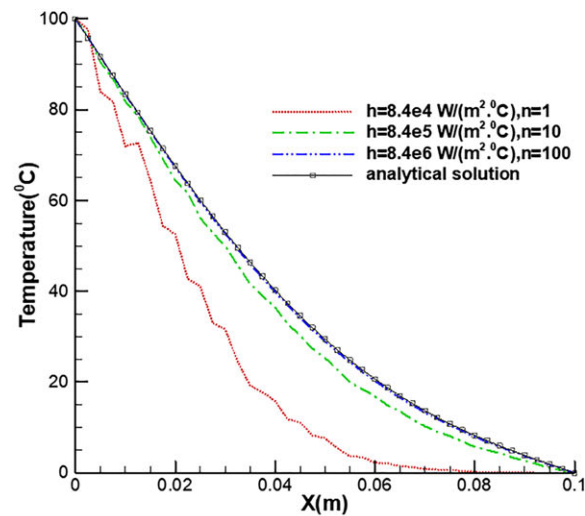
Keeping the thermal conductivity coefficient k of the tetrahedral element unchanged, we then alter the element size from $L_e = 0.005$ to 0.001 m, as shown in Figure 12. The numerical solution obtained by the 3D heat transfer model at $t = 20$ s is shown in Figure 13. When the element size L_e decreases from 0.005 to 0.001 m, the heat exchange



(A) $k = 4.2 \text{ W/m}^\circ\text{C}$, $L_e = 0.5 \times 10^{-2} \text{ m}$, $t = 200 \text{ s}$



(B) $k = 42 \text{ W/m}^\circ\text{C}$, $L_e = 0.5 \times 10^{-2} \text{ m}$, $t = 40 \text{ s}$



(C) $k = 420 \text{ W/m}^\circ\text{C}$, $L_e = 0.5 \times 10^{-2} \text{ m}$, $t = 6 \text{ s}$

FIGURE 11 Effect of the thermal conductivity coefficient on the heat exchange coefficient of the joint element. A, $k = 4.2 \text{ W/m}^\circ\text{C}$, $L_e = 0.5 \times 10^{-2} \text{ m}$, $t = 200 \text{ s}$; B, $k = 42 \text{ W/m}^\circ\text{C}$, $L_e = 0.5 \times 10^{-2} \text{ m}$, $t = 40 \text{ s}$; C, $k = 420 \text{ W/m}^\circ\text{C}$, $L_e = 0.5 \times 10^{-2} \text{ m}$, $t = 6 \text{ s}$ [Colour figure can be viewed at wileyonlinelibrary.com]

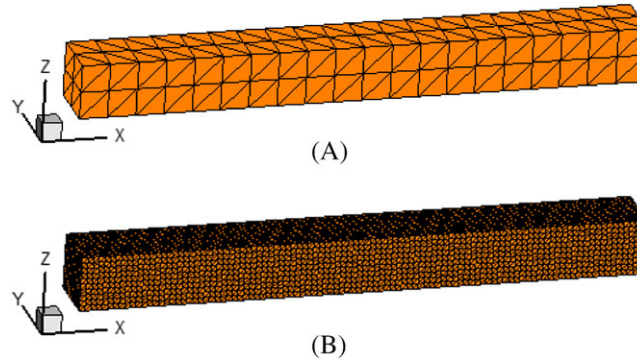


FIGURE 12 Computational mesh with different element sizes. A, Element size $L_e = 0.005$ m; B, element size $L_e = 0.001$ m [Colour figure can be viewed at wileyonlinelibrary.com]

coefficient h_j of the joint element should increase from 8.4×10^4 to 4.2×10^5 W/(m²·°C) (see Figure 13A,B). However, we find that the amplification factor n does not need to increase with the decrease of the element size. As long as $n \geq 100$, the numerical results of the 3D heat transfer model will be consistent with the analytical solution.

5.1.3 | Effect of the element shape on the heat transfer calculation

We maintain the thermal conductivity coefficient of the tetrahedral element, the element size, and the magnification factor $n=100$ and then change the element shape, as shown in Figure 14. The temperature distributions in the two cases are obtained by the 3D heat transfer model as shown in Figure 15. Although the element shape changes, the temperature distribution obtained by the heat transfer model is consistent with the analytical solution, and the errors of the numerical result for both cases stay below 0.5%. Thus, the element shape has little effect on the numerical results of the 3D heat transfer model.

In summary, as long as $n \geq 100$, this 3D heat transfer model can be used to characterize the heat conduction in a continuum.

Finally, taking $n = 100$, $L_e = 0.005$ m, $k=4.2$ W/m·°C, and the temperature distribution obtained by the 3D heat transfer model shown in Figure 16, it can be seen that the numerical solution agrees well with the analytical solution. The temperature distribution at different times is shown in Figure 17. This example indicates that the 3D heat transfer model can handle the heat conduction problem of continuous media.

5.2 | Influence of cracks on heat transfer

5.2.1 | Heat transfer in a medium with double cracks

A rectangular plate with thickness $H = 0.01$ m, $W = 0.05$ m, and length $L = 0.1$ m has two cracks with length 0.03 m that are located at positions $x = 0.03$ m and 0.07 m and are not perpendicular to the xy plane. Therefore, this problem is a true three-dimensional case. The boundary conditions and calculation parameters are the same as in Section 4.1. The 3D heat transfer model is used to study this problem, and the effect of cracks on heat transfer is discussed. A horizontal monitoring line is set on the top surface of the rectangular plate and through the center of the top surface, as shown in Figure 18 at $t = 40$ s. The temperature evolution of the rectangular plate obtained by the 3D heat transfer model (the heat transfer reduction factor of the broken joint element $r_T = 0$) is shown in Figure 18.

As shown in Figure 18, the high temperature propagates from left to right since the temperature on the left boundary is higher than that on the right boundary. The temperature spreads evenly from left to right when no crack is encountered, as shown in Figure 18 at $t = 40$ s. However, the temperature distribution cloud map forms a bulge pointing toward the first crack when a high temperature is transmitted to the vicinity of the first crack, as shown in Figure 18 at $t = 100$ s. Subsequently, the temperature distribution appears intermittently at the first crack, as shown in Figure 18 at $t = 140$ s since heat can only be transmitted to the right through the upper and lower sides of the first crack. However, because of the hindering effect of the crack on the heat transfer, the temperature on the right side of the first

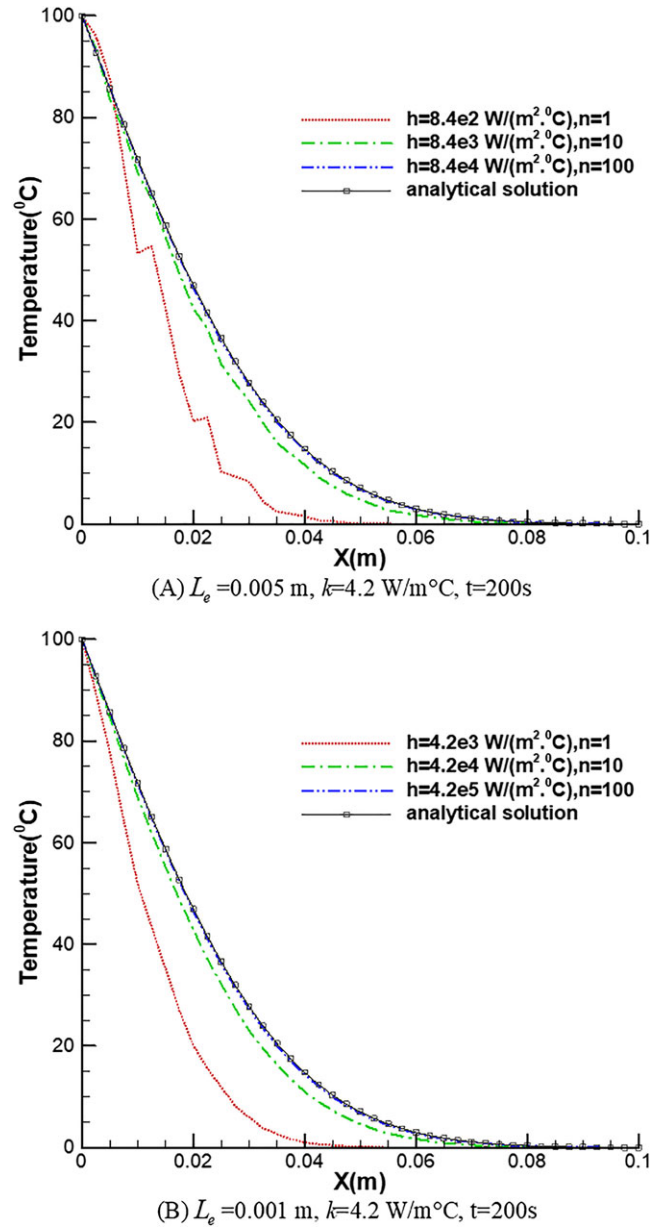


FIGURE 13 Effect of the element size on the heat exchange coefficient of the joint element. A, $L_e = 0.005$ m, $k = 4.2$ W/m°C, $t = 200$ s; B, $L_e = 0.001$ m, $k = 4.2$ W/m°C, $t = 200$ s [Colour figure can be viewed at wileyonlinelibrary.com]

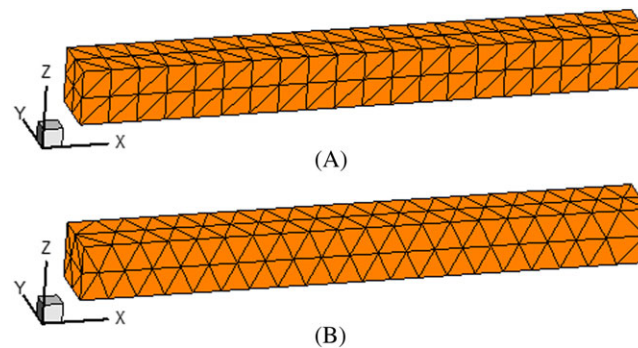


FIGURE 14 Computational mesh with different element shapes. A, Element shape 1; B, element shape 2 [Colour figure can be viewed at wileyonlinelibrary.com]

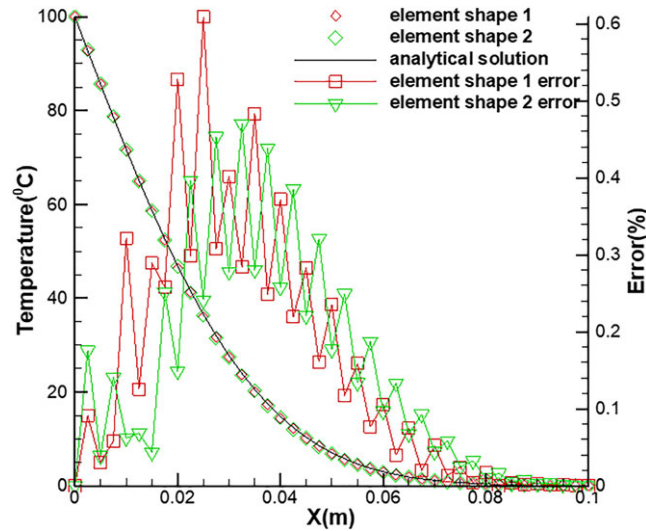


FIGURE 15 Temperature distribution for different element shapes at $t = 200$ s, $n = 100$ [Colour figure can be viewed at wileyonlinelibrary.com]

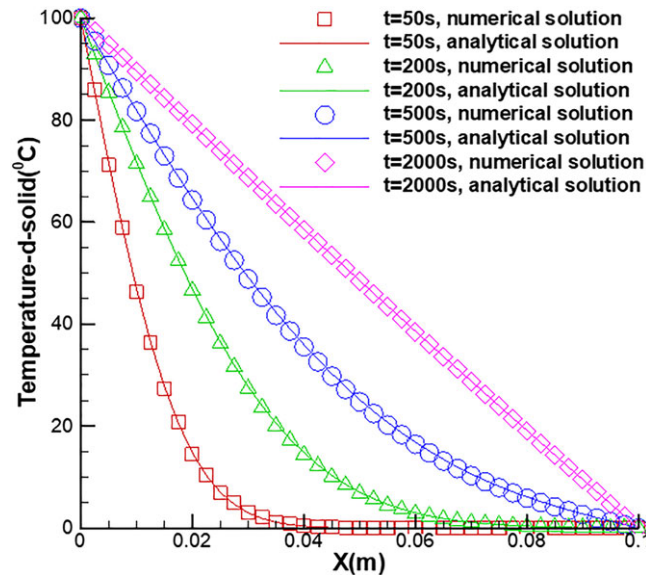


FIGURE 16 Numerical and analytical solutions of the temperature distribution in the block ($n = 100$) [Colour figure can be viewed at wileyonlinelibrary.com]

crack is always lower than the temperature on the left side of the first crack. Eventually, a sunken zone pointing toward the first crack forms on the right side of the first crack, as shown in Figure 18 at $t = 180$ s. Then the high temperature continues to propagate from left to right until the second crack is encountered. Similarly, a bulge that points toward the left side of the second crack forms, as shown in Figure 18 at $t = 300$ s. Because of the hindering effect of the second crack on the heat transfer, the temperature on the right side of the second crack is always lower than the temperature on the left side, and a sunken zone pointing toward the second crack also forms on the right side of the second crack, as shown at $t = 600$ s in Figure 18. After the heat transfer is stabilized, the temperature distribution is as shown at $t = 1200$ s in Figure 18.

To study the influence of the reduction factor r_T of the broken joint element on the heat transfer, we calculate the heat transfer process when the reduction factor r_T is 0, 0.001, 0.01, 0.1, and 1 using the 3D heat transfer model. The temperature distribution curve at $t = 500$ s is shown in Figure 19, where the analytical solution is the temperature distribution of the rectangular plate when no crack exists in it. As shown in Figure 19, the temperature jump amplitude at the two

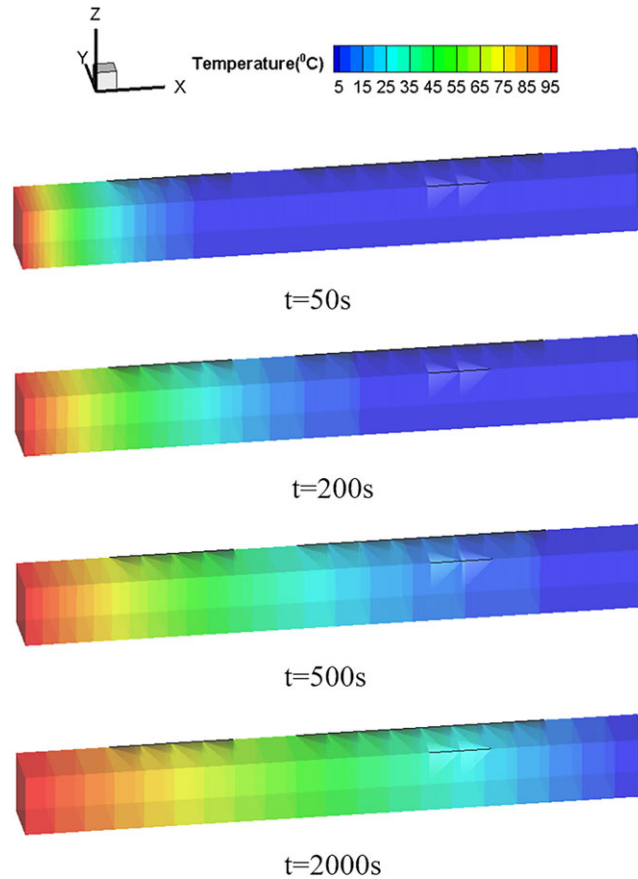


FIGURE 17 Temperature distribution in the block at different times ($n = 100$) [Colour figure can be viewed at wileyonlinelibrary.com]

cracks is very large if the reduction factor r_T is equal to 0. When the reduction factor gradually increases, the temperature jump amplitude at the two cracks gradually decreases, and the numerical solution gradually approaches the analytical solution. When $r_T = 1$, cracks do not affect the heat transfer, ie, this case can be considered as heat transfer in a continuous medium. Therefore, the numerical solution agrees well with the analytical solution when $r_T = 1$.

5.2.2 | Heat transfer in a medium with a complex fracture network

The model size, calculation parameters, and boundary conditions are the same as for the model in Section 5.2.1. However, there are multiple cracks in the rectangular plate. The 3D heat transfer model is also used to study the effect of multiple cracks on heat transfer. A horizontal monitoring line is set on the top surface of the rectangular plate and through the center of the top surface, as shown in Figure 21 for $r_T = 0$.

Figure 20 is the temperature distribution along the horizontal monitoring line when the reduction factor r_T of the joint element is 0, 0.001, 0.01, 0.1, and 1. When $r_T = 0$, the heat cannot pass through the cracks directly. The jump amplitude of the temperature at the cracks is very large, as shown in Figure 20 for $r_T = 0$. With an increase of reduction factor r_T , the temperature jump amplitude decreases because the heat can be directly transmitted from left to right through the cracks. However, the temperature will still mutate at the position where the horizontal monitoring line crosses the cracks, as shown in Figure 20 for $r_T = 0.001, 0.01$, and 0.1. The temperature distribution of the monitoring line gradually converges to the analytical solution with the increase of r_T . For $r_T = 1$, the temperature jump amplitude is 0, and the temperature distribution is consistent with the analytical solution.

The temperature distribution of the rectangular plate at $t = 500$ s is shown in Figure 21. Taking $r_T = 0$ in Figure 21 as an example, the cracks are fully insulated when $r_T = 0$, so heat can only be transferred to the right through the lower right continuous media. In Figure 21 when $r_T = 1$, the cracks do not affect the heat conduction, and the temperature gradually decreases from left to right. The distribution is very even and regular. The effects of the reduction factor r_T on the heat transfer can be clearly understood by combining Figure 20 and Figure 21.

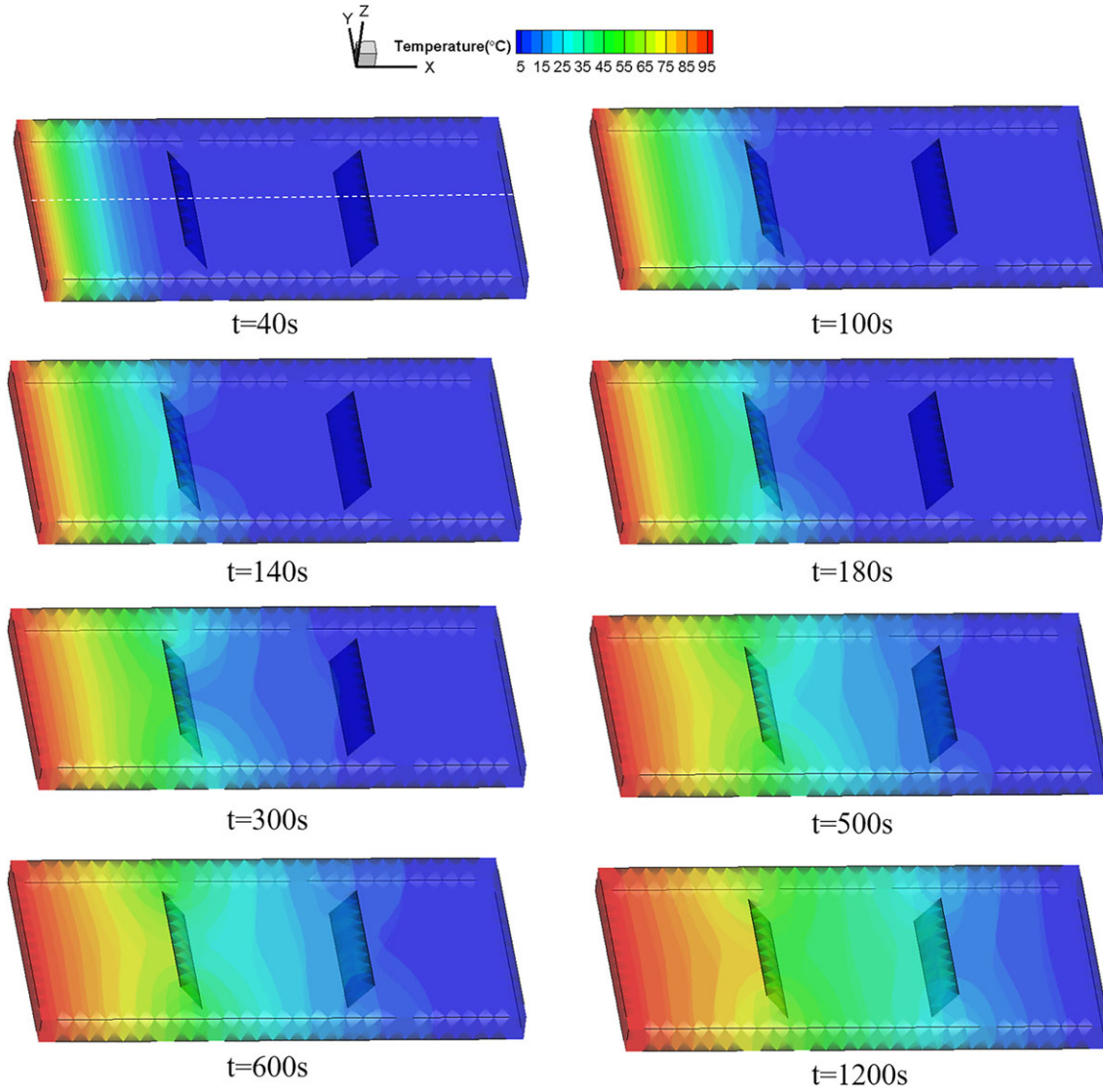


FIGURE 18 Temperature evolution in a medium with double cracks ($r_T = 0$) [Colour figure can be viewed at wileyonlinelibrary.com]

5.3 | Thermal stress calculation and thermally induced deformation

Take a rectangular plate with a length of $L = 1$ m, a width $W = 0.5$ m, and a thickness $H = 0.1$ m. The top and bottom boundaries are fixed, but the left, right, front, and rear boundaries are free. The initial temperature of the rectangular plate is T_0 . Assuming that the temperature of the rectangular plate increases by 100°C ($\Delta T = 100^\circ\text{C}$), the rectangular plate will undergo expansion deformation. However, because of the constraint of the top and bottom boundaries, the expansion deformation along the z direction is constrained, and thermal stress will be generated in the rectangular plate. Since along the x and y directions, the rectangular plate can be freely deformed, it will elongate along the x and y directions. The analytical solution of this problem is given as follows.

The total extension lengths along the x and y directions are given by

$$\begin{cases} \Delta L_x = L \left(-\frac{\mu \sigma_z}{E} + \alpha \Delta T \right) \\ \Delta L_y = W \left(-\frac{\mu \sigma_z}{E} + \alpha \Delta T \right) \end{cases} \quad (53)$$

The stress along the z direction is given by

$$\sigma_{zz} = E \alpha \Delta T. \quad (54)$$

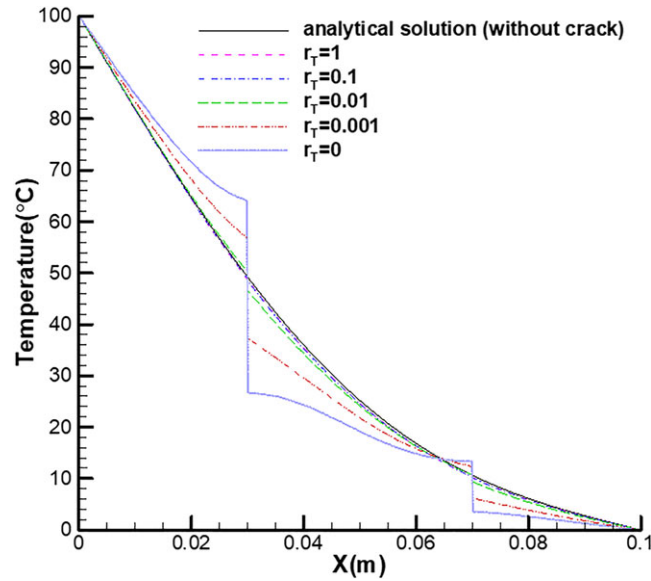


FIGURE 19 Temperature distribution on the horizontal monitoring line at $t = 500$ s [Colour figure can be viewed at wileyonlinelibrary.com]

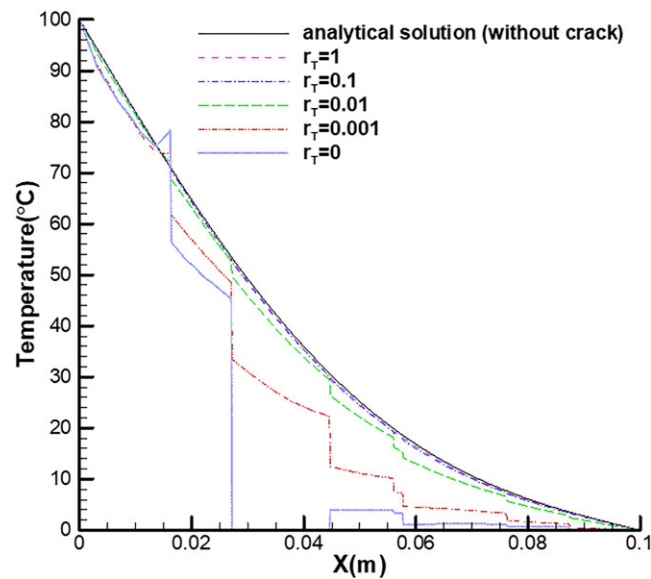


FIGURE 20 Temperature distribution along the horizontal monitoring line at $t = 500$ s for different reduction factors r_T [Colour figure can be viewed at wileyonlinelibrary.com]

The rectangular plate is discretized into 3575 tetrahedral elements with an element size of 0.05 m. The time step is 1×10^{-8} s. The calculation parameters include the elastic modulus $E = 20$ GPa, the Poisson ratio $\nu = 0.3$, and the thermal expansion coefficient $\alpha = 5 \times 10^{-5}$ ($1/^\circ\text{C}$).

The displacement vectors obtained by this 3D thermo-mechanical coupling model are shown in Figure 22, and the displacement distributions along the x and y directions are shown in Figure 23. It can be seen that the rectangular plate expands along the x and y directions, in which the displacement at the center is zero, and the displacement from the center to the outside gradually increases.

In addition, the stress along the z direction $\sigma_{zz} = 99.6$ MPa, and the error of the numerical result is 0.4% (the analytical solution is 100 MPa). The extension lengths along the x and y directions obtained from this thermo-mechanical model are 0.006516 and 0.0032592 m (the analytical solutions are 0.0065 and 0.00325 m), respectively. The errors are 0.25% and 0.28%, respectively. The numerical solution agrees well with the analytical solution. Thus, the accuracy of the 3D thermo-mechanical coupling model in solving thermally induced deformation and stress problems is verified.

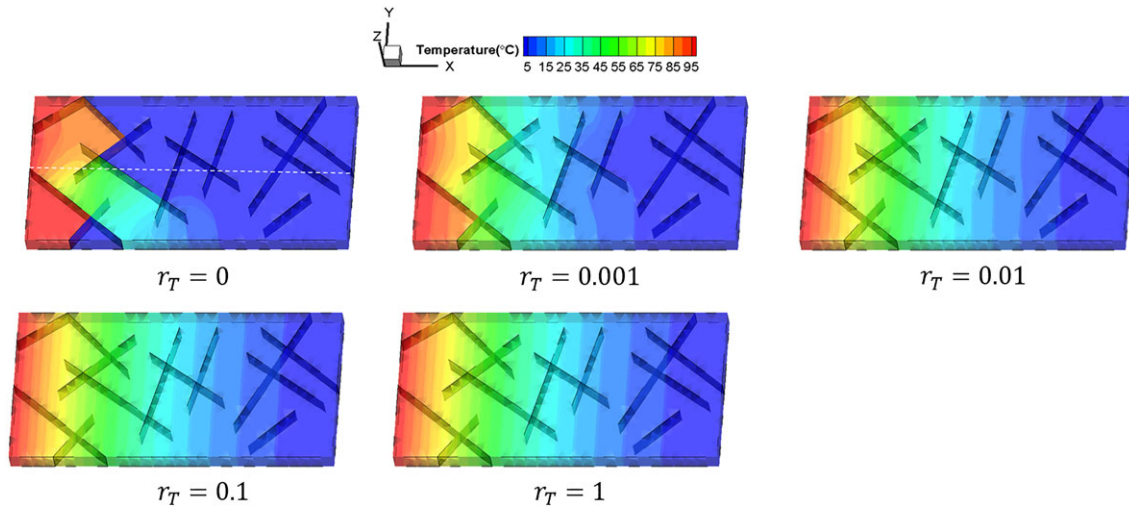


FIGURE 21 Temperature distribution along the monitoring line at $t = 500$ s for different reduction factors r_T of the broken joint element [Colour figure can be viewed at wileyonlinelibrary.com]

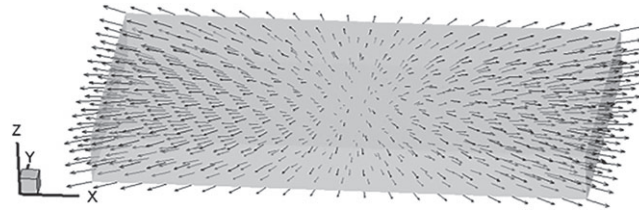


FIGURE 22 Displacement vectors

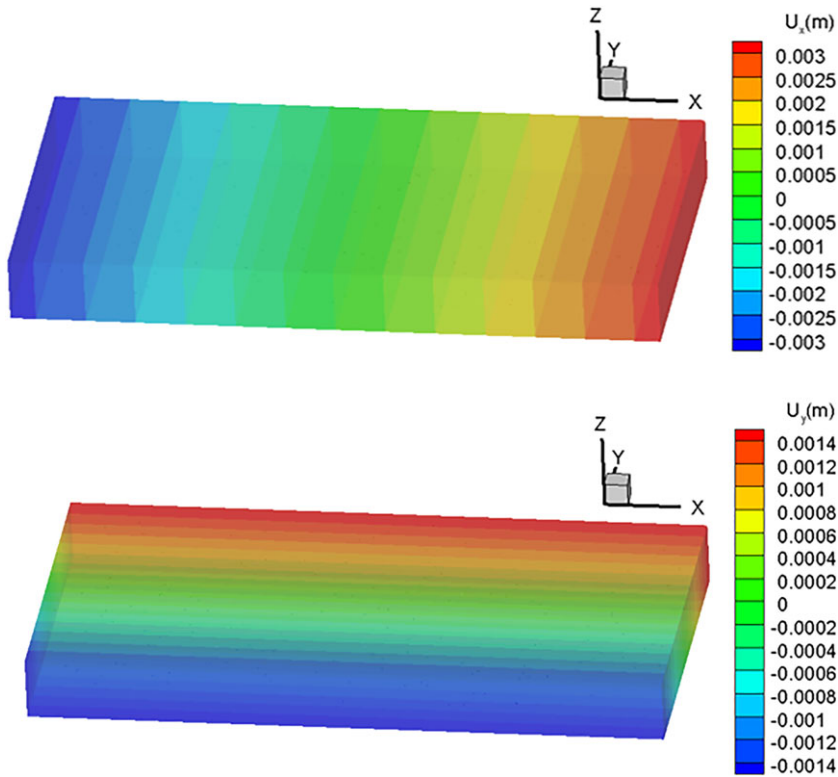


FIGURE 23 Displacement distribution along the x and y directions [Colour figure can be viewed at wileyonlinelibrary.com]

6 | THERMAL CRACKING EXPERIMENT

A granite specimen 22 cm in length and 30 cm in diameter was provided by the Canadian Underground Research Laboratory.¹² A hole with a diameter of 3 cm was drilled through the center of the granite specimen, as shown in

FIGURE 24 LdB marble specimen heating experiment,¹² and the numerical model [Colour figure can be viewed at wileyonlinelibrary.com]

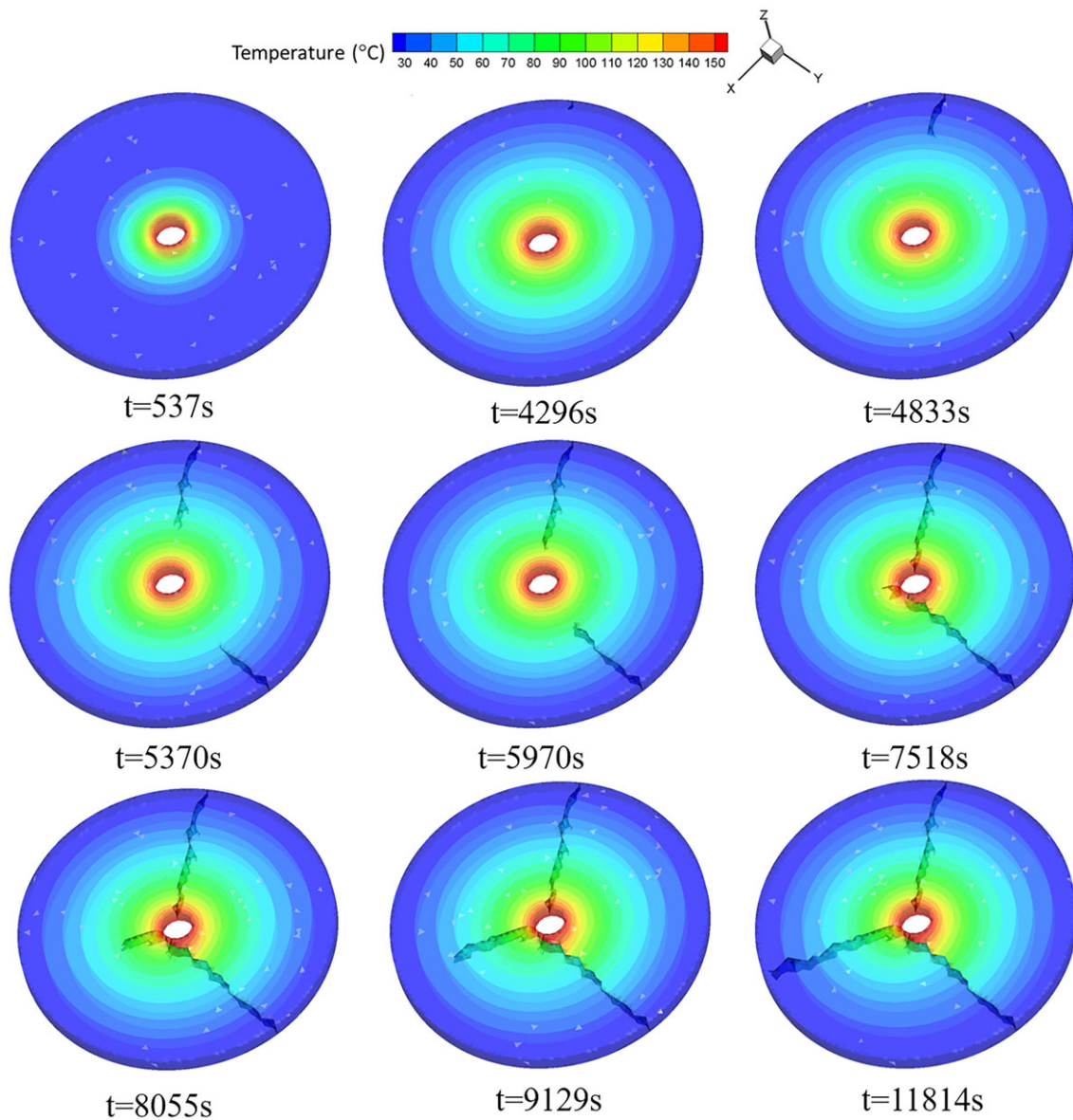
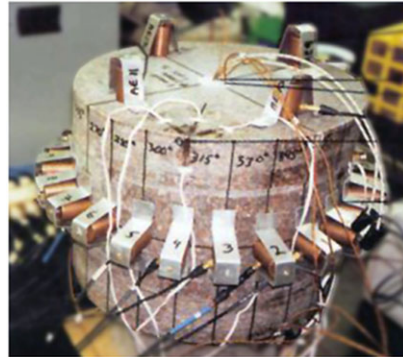


FIGURE 25 Thermal cracking process and temperature evolution of the granite specimen [Colour figure can be viewed at wileyonlinelibrary.com]

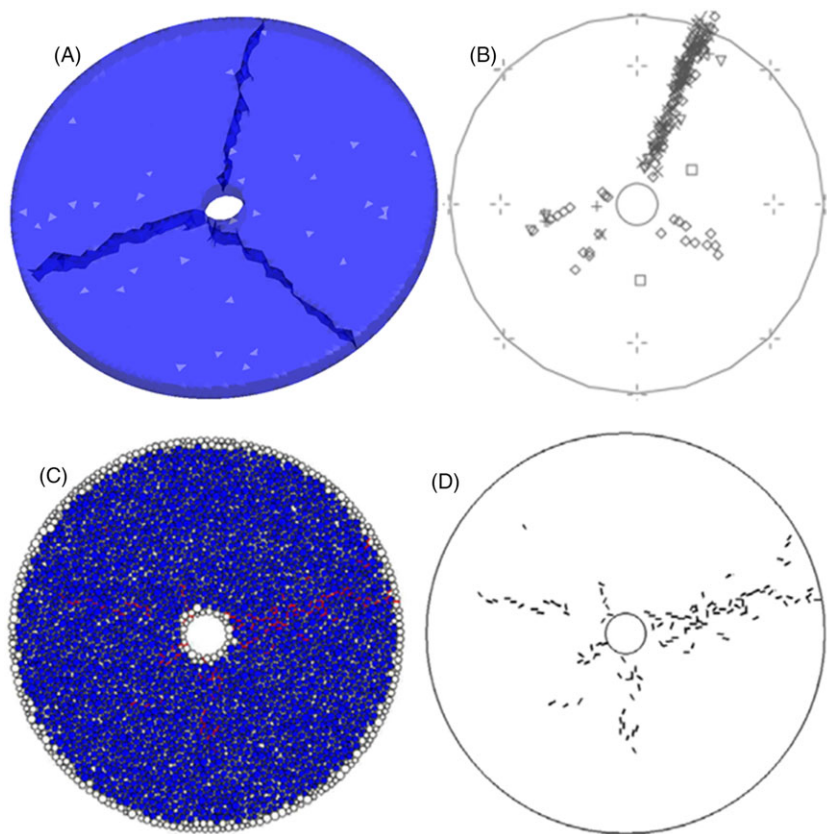


FIGURE 26 Comparison of the final crack morphology of cylindrical specimens: A, results obtained by the 3D thermo-mechanical coupling model; B, experimental results⁵²; C and D, results obtained by particle flow code¹³ [Colour figure can be viewed at wileyonlinelibrary.com]

Figure 24. The cylindrical granite specimen was heated via a heater inserted in the center hole, which finally caused the sample to undergo thermal cracking.

The 3D thermo-mechanical coupling model was used to simulate the experiment to verify the correctness of the model for simulating thermal cracking. To reduce the amount of calculations, we take the length of the numerical sample to be 1.8 cm. The initial temperature of the sample is 20°C. The maximum temperature of the heater is 160°C, and the temperature of the outer boundary of the sample is kept constant at 20°C. The mechanical and thermal parameters include the elastic modulus $E = 67$ GPa, the density $\rho = 2650$ kg/m³, Poisson ratio $\nu = 0.33$, the thermal conductivity coefficient $k = 3.5$ W/(m·°C), the specific heat $C_p = 1015$ J/(kg·°C), the thermal expansion coefficient $\alpha = 4.0 \times 10^{-6}$ 1/°C, and the heat exchange coefficient between the heater and rock $h_s = 1.0 \times 10^5$ W/(m²·°C).

The temperature evolution of the granite specimen and the thermal cracking process are shown in Figure 25. With an increase in temperature, cracks initiate from the outer boundary of the specimen, gradually extend to the inside of the specimen along the radial direction, and finally extend to the hole at the center of the sample, as shown in Figure 25 from $t = 4296$ s to $t = 11\,814$ s. The thermal cracking morphology obtained by the numerical model is similar to the fracture morphology observed in the laboratory⁵² and previous numerical simulation results,¹³ as shown in Figure 26. In addition, it can be clearly seen from $t = 9129$ s to $t = 11\,814$ s in Figure 24 that there is a significant discontinuity in the temperature distribution at the cracks, which indicates that the 3D thermo-mechanical coupling model can account for the hindering effect of the crack dynamic extension on the heat transfer. Some previous thermo-mechanical coupling models usually ignore the hindering effect of cracks on heat transfer.⁴⁰

7 | CONCLUSION

This paper presents a simple three-dimensional heat transfer model that can accurately account for the hindering effect of cracks on heat transfer. In addition, the criterion for selecting the heat exchange coefficient of the joint element is proposed. When the heat exchange coefficient of the joint element satisfies the criterion, the numerical result obtained by the 3D heat transfer model agrees well with the analytical solution. Moreover, a 3D thermo-mechanical coupling model is constructed to simulate the thermal cracking of rocks by combining the model with the mechanical fracture

calculation of the combined FDEM. The 3D heat transfer model has a wide range of applicability. In principle, it can also be applied to numerical methods such as the DEM, DDA, the NMM, and the FEM to construct thermo-mechanical coupling models that allow these methods to solve thermal cracking problems and dynamically consider the hindering effect of cracks on heat transfer.

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